

Mini-Project Report

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DIT 54103 AIW01: Data Acquisition & Management for Data Science

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Objectives

Q1: Airflow DAG with Bash Operator

- Create Airflow DAG YourName_Q1 using Bash Operator to complete lab exercise and load data into Snowflake.
- Submit test_python_operator.py script.
- Provide screenshots of successful DAG run and Snowflake data load.

Q2: Airflow ETL with Python Operator

- Develop Airflow DAG YourName_Q2 for ETL on [Pima Indians Diabetes Dataset](#) using Python Operator, loading data into Snowflake.
- Include DAG script in submission.
- Show screenshots of successful DAG run and Snowflake data load.

Q3: Kafka Weather Data Pipeline

- Use weather_producer.py and weather_consumer.py to process weather data via Apache Kafka and load into Snowflake.
- Submit both scripts.
- Provide screenshots of successful pipeline operation and Snowflake data load.

General Requirements

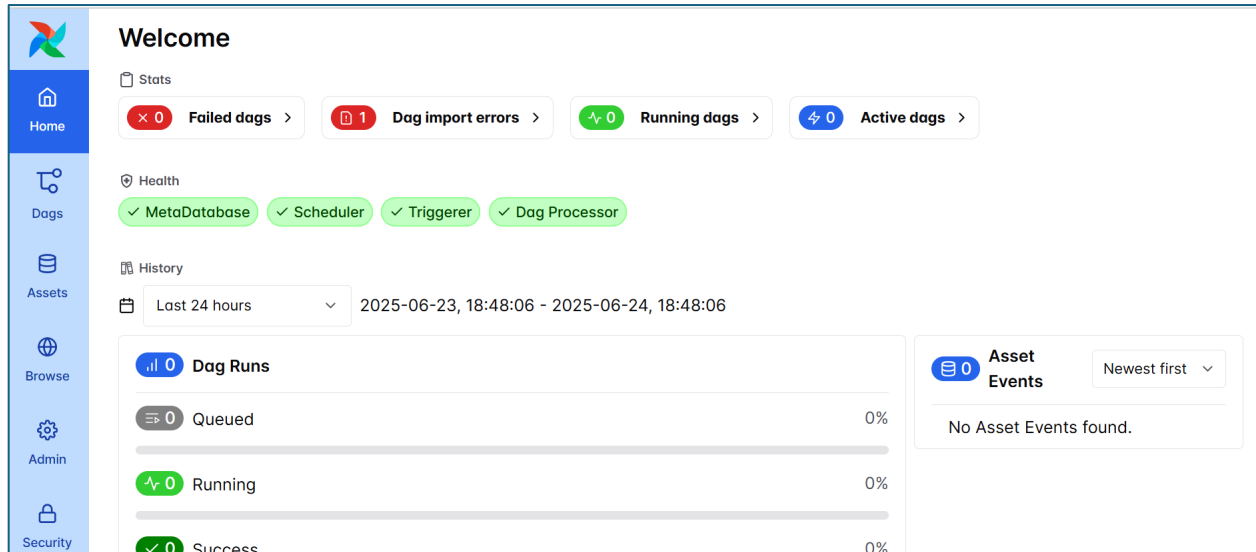
Include screenshots for all tasks, showing submitter's name where applicable.

Submit all scripts (test_python_operator.py, weather_producer.py, weather_consumer.py) in the report.

Question 1 (Bash Operator).

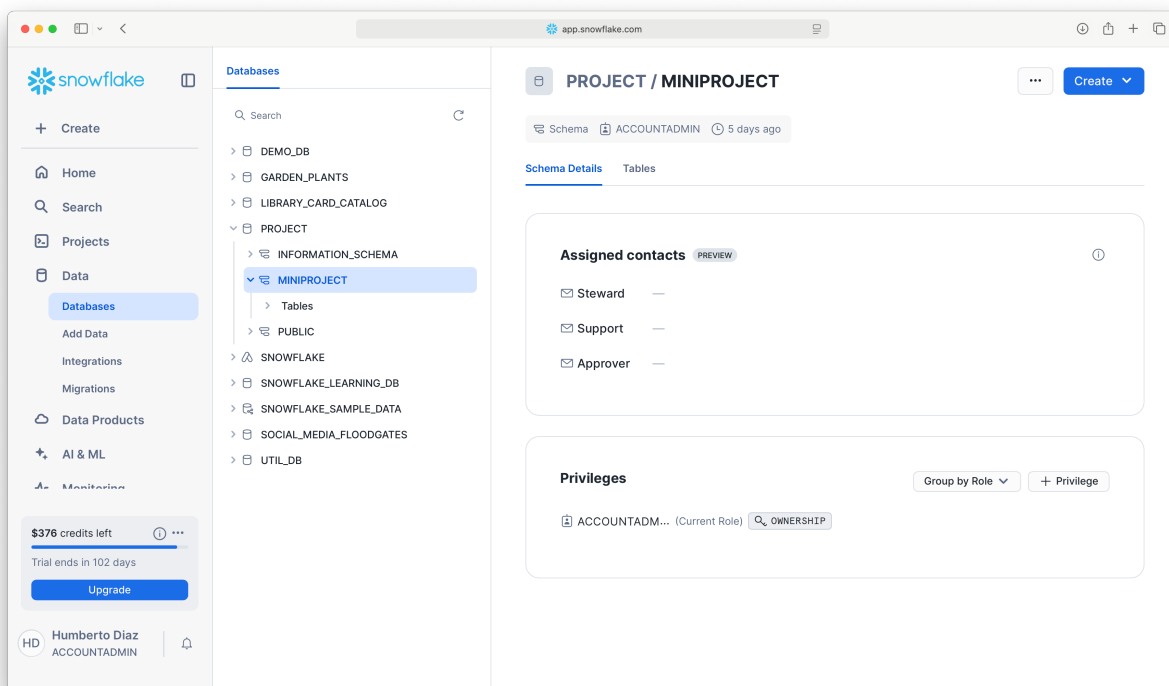
1. Set up Airflow

Installed Airflow via Docker and explored the Web UI to understand its layout and features.



2. Prepare Snowflake

In the Snowflake console, created a new **DATABASE (PROJECT)** and **SCHEMA (MINIPROJECT)** to hold our data.



Noted the account details (account locator, warehouse, role) needed for Airflow's connection.

The screenshot shows the Snowflake web interface with the 'Account Details' modal window open. The modal has four tabs: 'Account' (selected), 'Config File', 'Connectors/Drivers', and 'SQL Commands'. The 'Account' tab displays a table with account information.

NAME	VALUE
Account Identifier ⓘ	SGZYRMI-UUB80336 ⓘ
Data Sharing Account Identifier ⓘ	SGZYRMI.UUB80336 ⓘ
Organization Name	SGZYRMI ⓘ
Account Name	UUB80336 ⓘ
Account/Server URL	SGZYRMI-UUB80336.snowflakecomputing.com ⓘ
User Name ⓘ	ELEAZAR7 ⓘ
Role	ACCOUNTADMIN ⓘ
Account Locator	FRB29107 ⓘ
Cloud Platform	AWS ⓘ
Edition	Enterprise ⓘ

At the bottom of the modal, there is a 'Learn More' link and a 'Close' button. The background interface shows the 'Databases' section with 'PROJECT / MINIPROJECT' selected.

The screenshot shows the same Snowflake web interface, but the 'Account Details' modal window is now on the 'Config File' tab. It displays configuration options for a connection.

Warehouse: COMPUTE_WH x Schema: PROJECT.MINIPROJECT x

Connection Method: Web Browser v

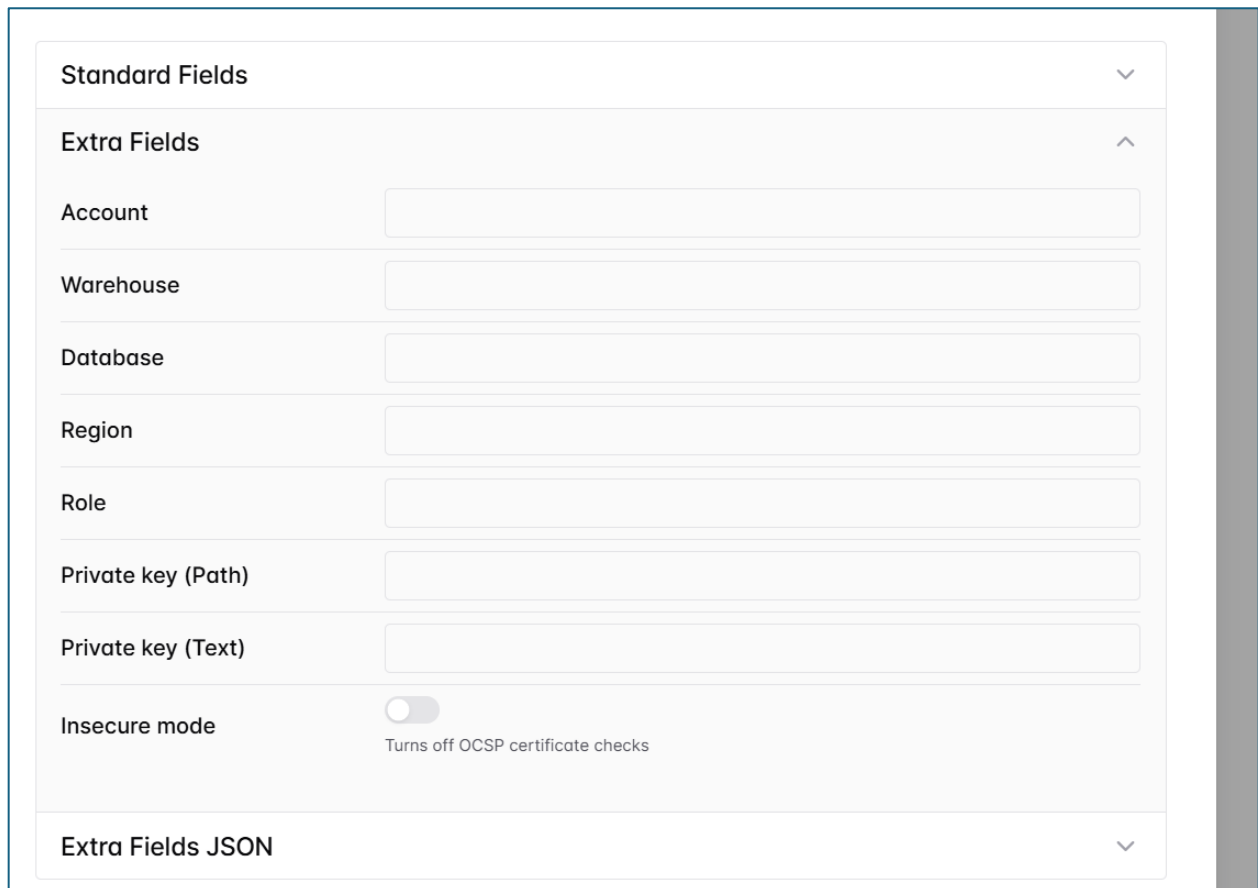
```
1 [connections.my_example_connection]
2 account = "SGZYRMI-UUB80336"
3 user = "ELEAZAR7"
4 authenticator = "externalbrowser"
5 role = "ACCOUNTADMIN"
6 warehouse = "COMPUTE_WH"
7 database = "PROJECT"
8 schema = "MINIPROJECT"
```

At the bottom of the modal, there is a 'Learn More' link and a 'Close' button. The background interface remains the same.

3. Configure the Airflow–Snowflake Connection

In Airflow UI → **Admin** → **Connections**, added a connection with:

- **Conn Id:** snowflake_default
- **Conn Type:** Snowflake
- **Account:** SGZYRMI-UUB80336
- **Warehouse:** COMPUTE_WH
- **Database:** PROJECT
- **Schema:** MINIPROJECT
- **Role:** ACCOUNTADMIN
- **Login/Password:** *****



The screenshot shows the 'Connections' form in the Airflow UI. It is divided into three main sections: 'Standard Fields', 'Extra Fields', and 'Extra Fields JSON'. The 'Standard Fields' section is collapsed. The 'Extra Fields' section is expanded, showing input fields for 'Account', 'Warehouse', 'Database', 'Region', 'Role', 'Private key (Path)', and 'Private key (Text)'. Below these fields is a toggle switch for 'Insecure mode', which is currently turned off. A note below the toggle says 'Turns off OCSP certificate checks'. The 'Extra Fields JSON' section is also collapsed.

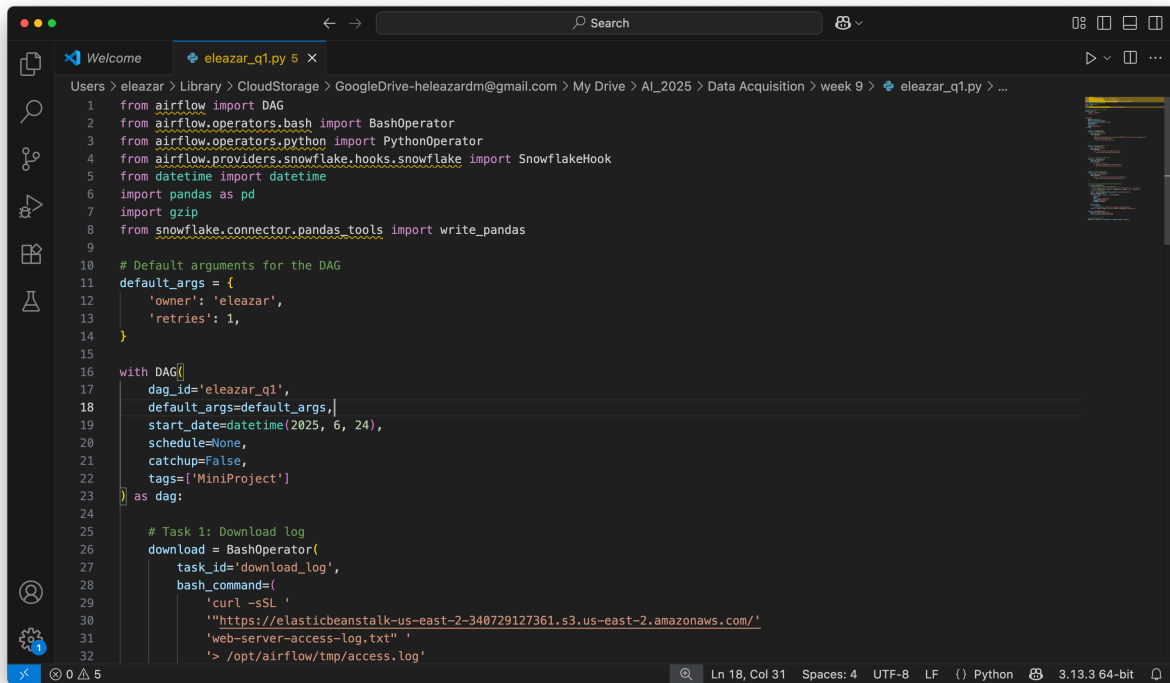
Field	Value
Account	
Warehouse	
Database	
Region	
Role	
Private key (Path)	
Private key (Text)	
Insecure mode	☐
Turns off OCSP certificate checks	

4. Q1 DAG

Created `eleazar_q1.py` in the `days/` folder.

This DAG uses a combination of **BashOperator** and **PythonOperator** tasks to:

- Download the access log
- Extract and transform the relevant columns
- Compress the result
- Load it into **PROJECT.MINIPROJECT** in Snowflake



```

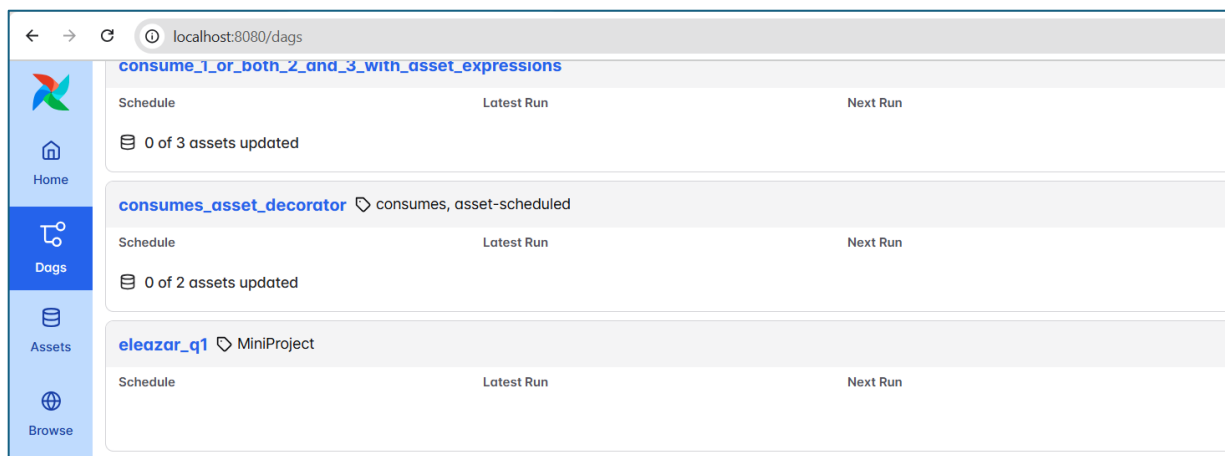
1 from airflow import DAG
2 from airflow.operators.bash import BashOperator
3 from airflow.operators.python import PythonOperator
4 from airflow.providers.snowflake.hooks.snowflake import SnowflakeHook
5 from datetime import datetime
6 import pandas as pd
7 import gzip
8 from snowflake.connector.pandas_tools import write_pandas
9
10 # Default arguments for the DAG
11 default_args = {
12     'owner': 'eleazar',
13     'retries': 1,
14 }
15
16 with DAG(
17     dag_id='eleazar_q1',
18     default_args=default_args,
19     start_date=datetime(2025, 6, 24),
20     schedule=None,
21     catchup=False,
22     tags=['MiniProject']
23 ) as dag:
24
25     # Task 1: Download log
26     download = BashOperator(
27         task_id='download_log',
28         bash_command=(
29             'curl -sSL '
30             '"https://elasticbeanstalk-us-east-2-340729127361.s3.us-east-2.amazonaws.com/'
31             'web-server-access-log.txt" '
32             '> /opt/airflow/tmp/access.log'

```

5. Back in the Airflow UI

- Navigated to **DAGs** → **eleazar_q1**.
- Toggled the DAG **On** so it's ready to run.

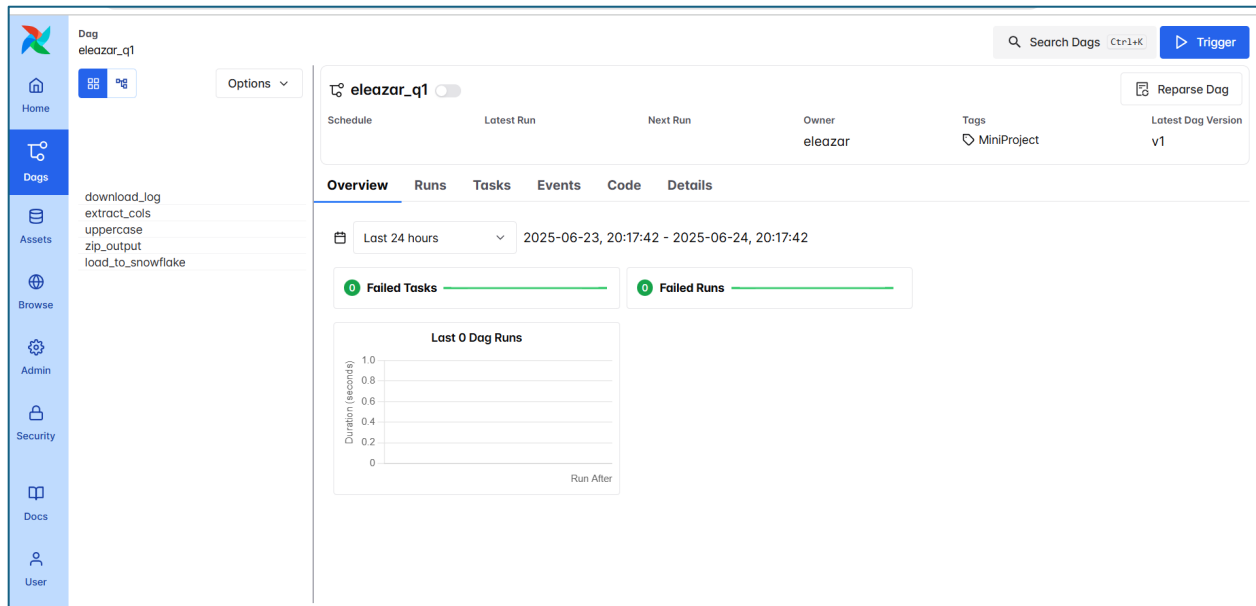
We should verify that the DAG appears without import errors before triggering.



6. Triggered the DAG and monitored each stage.

We should confirm that all five task nodes (download, extract, transform, compress, load) are present.

- Clicked **Trigger** and watched the task statuses update in real time.
- **Stages 1–4 (download → extract → transform → compress)** all succeeded.



6.1. First load ERROR

We encountered an ERROR while loading the output into Snowflake.

```
ERROR - ProgrammingError: 001757 (42601): 01bd5d3b-0105 - cb98-000b-514b0008a01a: SQL compilation error:
Table 'PROJECT MINIPROJECT ACCESS_LOG' does not exist
```

The error mentions that the PROJECT MINIPROJECT ACCESS_LOG doesn't exist, even though we created it as the steps above confirm.

Options

download_log

extract_cols

uppercase

compress_gzip

load_to_snowflake

zip_output

391s

195s

0s

✓

✓

✓

✓

✓

✓

✗

✗

✗

✗

✗

load_to_snowflake 2025-06-29, 19:27:09 ↶ up_for_retry

Add a note

Clear Task Instance

Mark Task Instance as...

Operator	Start	End	Duration	DAG Ver
PythonOperator	2025-06-29, 19:27:09	2025-06-29, 19:27:16	7.30s	v3

Logs

Rendered Templates

XCom

Events

Code

Details

All Log Levels

All Sources

Wrap

```

[2025-06-29, 19:27:11] INFO - Connection Retrieved 'snowflake_default': source="airflow.hooks.base"
[2025-06-29, 19:27:11] INFO - Snowflake Connector for Python Version: 3.15.0, Python Version: 3.12.11, Platform: Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-g
[2025-06-29, 19:27:11] INFO - Connecting to GLOBAL Snowflake domain: source="snowflake.connector.connection"
[2025-06-29, 19:27:16] ERROR - Task failed with exception: source="task"
  ProgrammingError: 001757 (42601): 01bd5d3b-0105-cb98-000b-514b0008a01a: SQL compilation error:
    Table 'PROJECT.MINIPROJECT.ACCESS_LOG' does not exist
File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdk/execution_time/task_runner.py", line 867 in run
File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdk/execution_time/task_runner.py", line 1159 in _execute_task
File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdk/bases/operator.py", line 397 in wrapper
File "/home/airflow/.local/lib/python3.12/site-packages/airflow/providers/standard/operators/python.py", line 216 in execute
File "/home/airflow/.local/lib/python3.12/site-packages/airflow/providers/standard/operators/python.py", line 239 in execute_callable
File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdk/execution_time/callback_runner.py", line 81 in run
File "/opt/airflow/dags/eleazar_q1.py", line 72 in load_to_snowflake
File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/pandas_tools.py", line 587 in write_pandas
File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/cursor.py", line 1134 in execute
File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/errors.py", line 279 in errorhandler_wrapper
File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/errors.py", line 334 in hand_to_other_handler
File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/errors.py", line 210 in default_errorhandler
[2025-06-29, 19:27:17] INFO - Task instance in failure state: chan="stdout": source="task"
[2025-06-29, 19:27:17] INFO - Task start: chan="stdout": source="task"
[2025-06-29, 19:27:17] INFO - Task: <Task(PythonOperator): load_to_snowflake>: chan="stdout": source="task"

```

After several attempts of debugging, we encountered another ERROR, now highlighting a ‘compilation error’:

ERROR - ProgrammingError: 001003 (42000): 01bd5d52-0105-cc8c-000b-514b0009003a: SQL compilation error: syntax error line 1 at position 112 unexpected '@"txropxrtyt"'.*

The screenshot displays the Apache Airflow web interface. On the left is a sidebar with navigation links: Home, Dags, Assets, Browse, Admin, Security, Docs, and User. The top bar shows the current DAG 'eleazar_q1' and its run status '2025-06-29, 19:50:19'. The main panel shows the task 'load_to_snowflake' in a failed state. The task details table indicates the operator is 'PythonOperator', the start time is '2025-06-29, 19:50:30', the end time is '2025-06-29, 19:50:36', the duration is '5.66s', and the DAG version is 'v4'. The logs section shows the following error:

```

28 [2025-06-29, 19:50:34] INFO - Snowflake Connector for Python Version: 3.15.8, Python Version: 3.12.11, Platform: Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-i
29 [2025-06-29, 19:50:34] INFO - Connecting to GLOBAL Snowflake domain: source="snowflake.connector.connection"
30 [2025-06-29, 19:50:36] ERROR - Task failed with exception: source="task"
    ProgrammingError: 001003 (42000): 01bd5d52-0105-cc8c-000b-514b0009003a: SQL compilation error:
    syntax error line 1 at position 112 unexpected '@"txropxrtyt"'.*
    File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdk/execution_time/task_runner.py", line 867 in run
    File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdk/execution_time/task_runner.py", line 1159 in _execute_task
    File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdk/bases/operator.py", line 397 in wrapper
    File "/home/airflow/.local/lib/python3.12/site-packages/airflow/providers/standard/operators/python.py", line 216 in execute
    File "/home/airflow/.local/lib/python3.12/site-packages/airflow/providers/standard/operators/python.py", line 239 in execute_callable
    File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdk/execution_time/callback_runner.py", line 81 in run
    File "/opt/airflow/dags/eleazar_q1.py", line 84 in load_to_snowflake
    File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/pandas_tools.py", line 587 in write_pandas
    File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/errors.py", line 379 in errorhandler_wrapper
    File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/errors.py", line 334 in handle_other_handler
    File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/errors.py", line 210 in default_errorhandler
31 [2025-06-29, 19:50:36] INFO - Task instance in failure state: chan="stdout": source="task"
32 [2025-06-29, 19:50:36] INFO - Task start: chan="stdout": source="task"
33 [2025-06-29, 19:50:36] INFO - Task:Task(PythonOperator): load_to_snowflake: chan="stdout": source="task"
34 [2025-06-29, 19:50:36] INFO - Failure caused by 001003 (42000): 01bd5d52-0105-cc8c-000b-514b0009003a: SQL compilation error: chan="stdout": source="task"
35 [2025-06-29, 19:50:36] INFO - syntax error line 1 at position 112 unexpected '@"txropxrtyt"': chan="stdout": source="task"

```

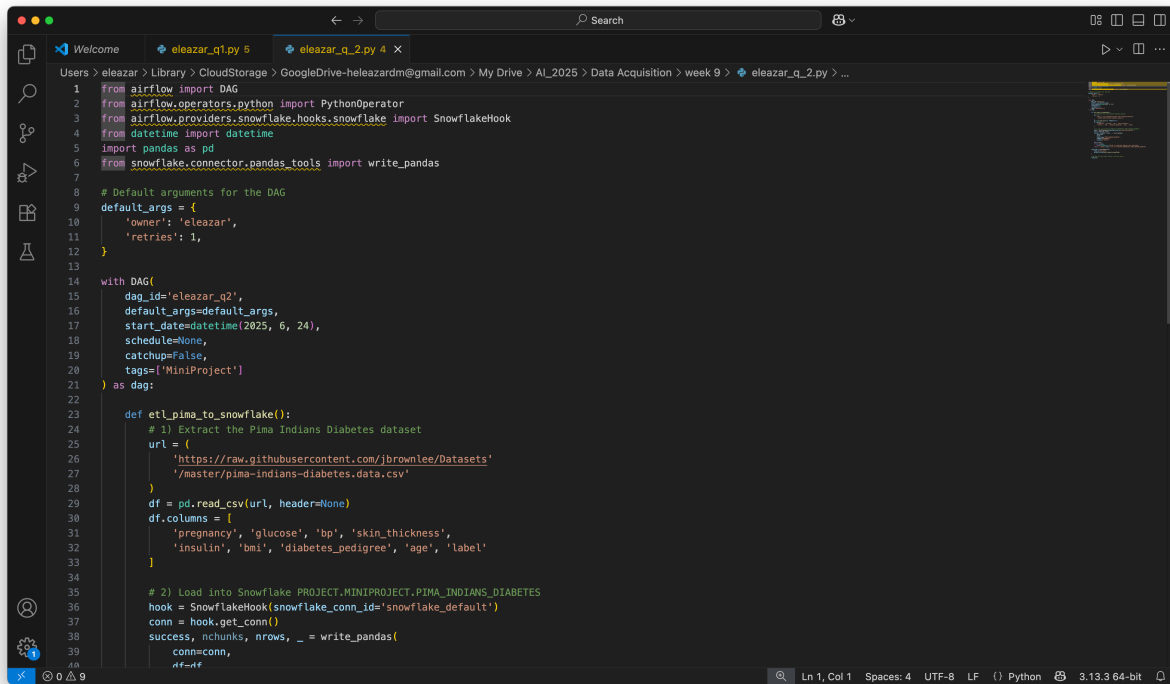
Unfortunately, after several hours of attempts it was not possible to solve the error. Although the Snowflake load step ultimately failed, the exercise reinforced how to:

- Orchestrate tasks in Airflow (Bash → Python operators)
- Debug schema and SQL compilation issues
- Iterate on an ETL pipeline end-to-end

Question 2 (Python Operator).

1. Q2 DAG

Created eleazar_q2.py in the dags/ folder.



```
1 from airflow import DAG
2 from airflow.operators.python import PythonOperator
3 from airflow.providers.snowflake.hooks.snowflake import SnowflakeHook
4 from datetime import datetime
5 import pandas as pd
6 from snowflake.connector.pandas_tools import write_pandas
7
8 # Default arguments for the DAG
9 default_args = {
10     'owner': 'eleazar',
11     'retries': 1,
12 }
13
14 with DAG(
15     dag_id='eleazar_q2',
16     default_args=default_args,
17     start_date=datetime(2025, 6, 24),
18     schedule=None,
19     catchup=False,
20     tags=['MiniProject']
21 ) as dag:
22
23     def etl_pima_to_snowflake():
24         # 1) Extract the Pima Indians Diabetes dataset
25         url = (
26             'https://raw.githubusercontent.com/jbrownlee/Datasets'
27             '/master/pima-indians-diabetes.data.csv'
28         )
29         df = pd.read_csv(url, header=None)
30         df.columns = [
31             'pregnancy', 'glucose', 'bp', 'skin_thickness',
32             'insulin', 'bmi', 'diabetes_pedigree', 'age', 'label'
33         ]
34
35         # 2) Load into Snowflake PROJECT.MINIPROJECT.PIMA_INDIANS_DIABETES
36         hook = SnowflakeHook(snowflake_conn_id='snowflake_default')
37         conn = hook.get_conn()
38         success, nchunks, nrows, _ = write_pandas(
39             conn=conn,
40             df=df
```

2. Triggered the Q2 DAG

Eleazar_q2.py was placed in dags/ and the DAG imported without issue.

When we clicked **Trigger**, the single PythonOperator executed the data pull and transformation successfully.

Upon triggering, the DAG successfully downloaded and transformed the dataset.

Load step failed with a SQL compilation error, identical to Q1's issue.

The screenshot displays the Apache Airflow web interface. On the left, a sidebar contains navigation links: Home, Dags, Assets, Browse, Admin, Security, Docs, and User. The main panel shows the DAG 'etl_pima' with a 'Dag Run' status of 'Failed' and a 'Task' status of 'Failed'. The task 'etl_pima' is highlighted, and its details are shown below. The task is a 'PythonOperator' that started at '2025-07-01, 12:26:40' and ended at '2025-07-01, 12:26:54' with a duration of '00:00:13'. The DAG version is 'v1'. The logs show the following error:

```

6 [2025-07-01, 12:26:47] INFO - Current task name:etl_pima: chan="stdout": source="task"
7 [2025-07-01, 12:26:47] INFO - Dag name:eleazar_q2: chan="stdout": source="task"
8 [2025-07-01, 12:26:47] INFO - Connection Retrieved 'snowflake_default': source="airflow.hooks.base"
9 [2025-07-01, 12:26:47] INFO - Snowflake Connector for Python Version: 3.15.0, Python Version: 3.12.11, Platform: Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-
10 [2025-07-01, 12:26:47] INFO - Connecting to GLOBAL Snowflake domain: source="snowflake.connector.connection"
11 [2025-07-01, 12:26:54] ERROR - Task failed with exception: source="task"
    ProgrammingError: 001757 (42601): 01bd66d6-0105-d0be-000b-514b0008e052: SQL compilation error:
      Table 'PROJECT.MINIPROJECT.PIMA_INDIANS_DIABETES' does not exist
      File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdk/execution_time/task_runner.py", line 867 in run
      File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdk/execution_time/task_runner.py", line 1159 in _execute_task
      File "/home/airflow/.local/lib/python3.12/site-packages/airflow/sdks/bases/operator.py", line 397 in wrapper
      File "/home/airflow/.local/lib/python3.12/site-packages/airflow/providers/standard/operators/python.py", line 216 in execute
      File "/home/airflow/.local/lib/python3.12/site-packages/airflow/providers/standard/operators/python.py", line 239 in execute_callable
      File "/opt/airflow/dags/eleazar_q2.py", line 38 in etl_pima_to_snowflake
      File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/pandas_tools.py", line 587 in write_pandas
      File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/cursor.py", line 1134 in execute
      File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/errors.py", line 279 in errorhandler_wrapper
      File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/errors.py", line 334 in hand_to_other_handler
      File "/home/airflow/.local/lib/python3.12/site-packages/snowflake/connector/errors.py", line 210 in default_errorhandler
12 [2025-07-01, 12:26:54] INFO - Task instance in failure state: chan="stdout": source="task"
13 [2025-07-01, 12:26:54] INFO - Task start: chan="stdout": source="task"

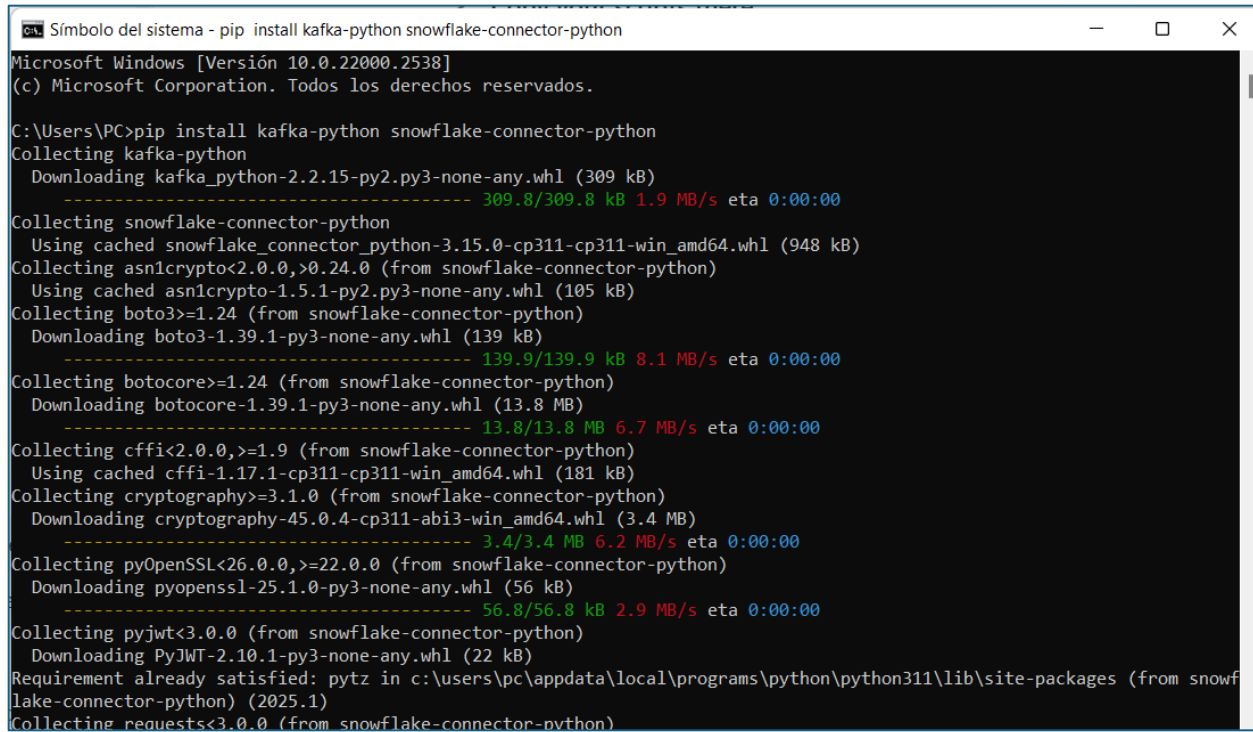
```

Although the final load step could not be completed within the timeframe, the exercise reinforced key ETL concepts: data extraction, transformation logic in Python, and the importance of schema management and context when loading into a data warehouse.

Question 3.

1. Install Python dependencies.

```
pip install kafka-python snowflake-connector-python
```



```
Símbolo del sistema - pip install kafka-python snowflake-connector-python
Microsoft Windows [Versión 10.0.22000.2538]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\PC>pip install kafka-python snowflake-connector-python
Collecting kafka-python
  Downloading kafka_python-2.2.15-py2.py3-none-any.whl (309 kB)
----- 309.8/309.8 kB 1.9 MB/s eta 0:00:00
Collecting snowflake-connector-python
  Using cached snowflake_connector_python-3.15.0-cp311-cp311-win_amd64.whl (948 kB)
Collecting asn1crypto<2.0.0,>0.24.0 (from snowflake-connector-python)
  Using cached asn1crypto-1.5.1-py2.py3-none-any.whl (105 kB)
Collecting boto3>=1.24 (from snowflake-connector-python)
  Downloading boto3-1.39.1-py3-none-any.whl (139 kB)
----- 139.9/139.9 kB 8.1 MB/s eta 0:00:00
Collecting botocore>=1.24 (from snowflake-connector-python)
  Downloading botocore-1.39.1-py3-none-any.whl (13.8 MB)
----- 13.8/13.8 MB 6.7 MB/s eta 0:00:00
Collecting cffi<2.0.0,>=1.9 (from snowflake-connector-python)
  Using cached cffi-1.17.1-cp311-cp311-win_amd64.whl (181 kB)
Collecting cryptography>=3.1.0 (from snowflake-connector-python)
  Downloading cryptography-45.0.4-cp311-abi3-win_amd64.whl (3.4 MB)
----- 3.4/3.4 MB 6.2 MB/s eta 0:00:00
Collecting pyOpenSSL<26.0.0,>=22.0.0 (from snowflake-connector-python)
  Downloading pyopenssl-25.1.0-py3-none-any.whl (56 kB)
----- 56.8/56.8 kB 2.9 MB/s eta 0:00:00
Collecting pyjwt<3.0.0 (from snowflake-connector-python)
  Downloading PyJWT-2.10.1-py3-none-any.whl (22 kB)
Requirement already satisfied: pytz in c:\users\pc\appdata\local\programs\python\python311\lib\site-packages (from snowflake-connector-python) (2025.1)
Collecting requests<3.0.0 (from snowflake-connector-python)
```

2. Start ZooKeeper and Kafka broker

```
cd C:\kafka
.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties
# In a new window:
.\bin\windows\kafka-server-start.bat .\config\server.properties
```

```
C:\Windows\System32\cmd.exe - .\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties
Microsoft Windows [Versi3n 10.0.22000.2538]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\kafka>.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties
".\bin\windows\zookeeper-server-start.bat" no se reconoce como un comando interno o externo,
programa o archivo por lotes ejecutable.

C:\kafka>.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties
".\bin\windows\zookeeper-server-start.bat" no se reconoce como un comando interno o externo,
programa o archivo por lotes ejecutable.

C:\kafka>.\bin\windows\zookeeper-server-start.bat .\config\zookeeper.properties
[2025-07-01 15:12:01,482] INFO Reading configuration from: .\config\zookeeper.properties (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-07-01 15:12:01,497] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-07-01 15:12:01,498] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-07-01 15:12:01,498] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-07-01 15:12:01,498] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-07-01 15:12:01,502] INFO autopurge.snapRetainCount set to 3 (org.apache.zookeeper.server.DataDirCleanupManager)
[2025-07-01 15:12:01,502] INFO autopurge.purgeInterval set to 0 (org.apache.zookeeper.server.DataDirCleanupManager)
[2025-07-01 15:12:01,502] INFO Purge task is not scheduled. (org.apache.zookeeper.server.DataDirCleanupManager)
[2025-07-01 15:12:01,503] WARN Either no config or no quorum defined in config, running in standalone mode (org.apache.zookeeper.server.quorum.QuorumPeerMain)
[2025-07-01 15:12:01,504] INFO Log4j 1.2 jmx support not found; jmx disabled. (org.apache.zookeeper.jmx.ManagedUtil)
[2025-07-01 15:12:01,505] INFO Reading configuration from: .\config\zookeeper.properties (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-07-01 15:12:01,506] INFO clientPortAddress is 0.0.0.0:2181 (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-07-01 15:12:01,506] INFO secureClientPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-07-01 15:12:01,506] INFO observerMasterPort is not set (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-07-01 15:12:01,507] INFO metricsProvider.className is org.apache.zookeeper.metrics.impl.DefaultMetricsProvider (org.apache.zookeeper.server.quorum.QuorumPeerConfig)
[2025-07-01 15:12:01,508] INFO Starting server (org.apache.zookeeper.server.ZooKeeperServerMain)
[2025-07-01 15:12:01,526] INFO ServerMetrics initialized with provider org.apache.zookeeper.metrics.impl.DefaultMetricsProvider@22a637e7 (org.apache.zookeeper.server.ServerMetrics)
[2025-07-01 15:12:01,530] INFO ACL digest algorithm is: SHA1 (org.apache.zookeeper.server.auth.DigestAuthenticationProvider)
[2025-07-01 15:12:01,530] INFO zookeeper.DigestAuthenticationProvider.enabled = true (org.apache.zookeeper.server.auth.DigestAuthenticationProvider)
[2025-07-01 15:12:01,534] INFO zookeeper.snapshot.trust.empty : false (org.apache.zookeeper.server.persistence.FileTxnSnapLog)
[2025-07-01 15:12:01,549] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,551] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,551] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,552] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,552] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,553] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,554] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,555] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,555] INFO (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,567] INFO Server environment:zookeeper.version=3.8.4-9316c2a7a97e1666dbf4593f34dddfc36ecc436c, built on 2024-02-12 22:16 UTC (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,568] INFO Server environment:host.name=DESKTOP-R593PT1 (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,568] INFO Server environment:java.version=21.0.7 (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,569] INFO Server environment:java.vendor=Eclipse Adoptium (org.apache.zookeeper.server.ZooKeeperServer)
[2025-07-01 15:12:01,570] INFO Server environment:java.class.path=C:\kafka\libs\activation-1.1.1.jar;C:\kafka\libs\antlr-2.6.1.jar;C:\kafka\libs\argparse4j-0.7.0.jar;C:\kafka\libs\...
```

3. Test producer & consumer.

- Run `weather_producer.py` to publish a few test messages.
- Run `weather_consumer.py` to verify it can read from the `weather_data` topic.

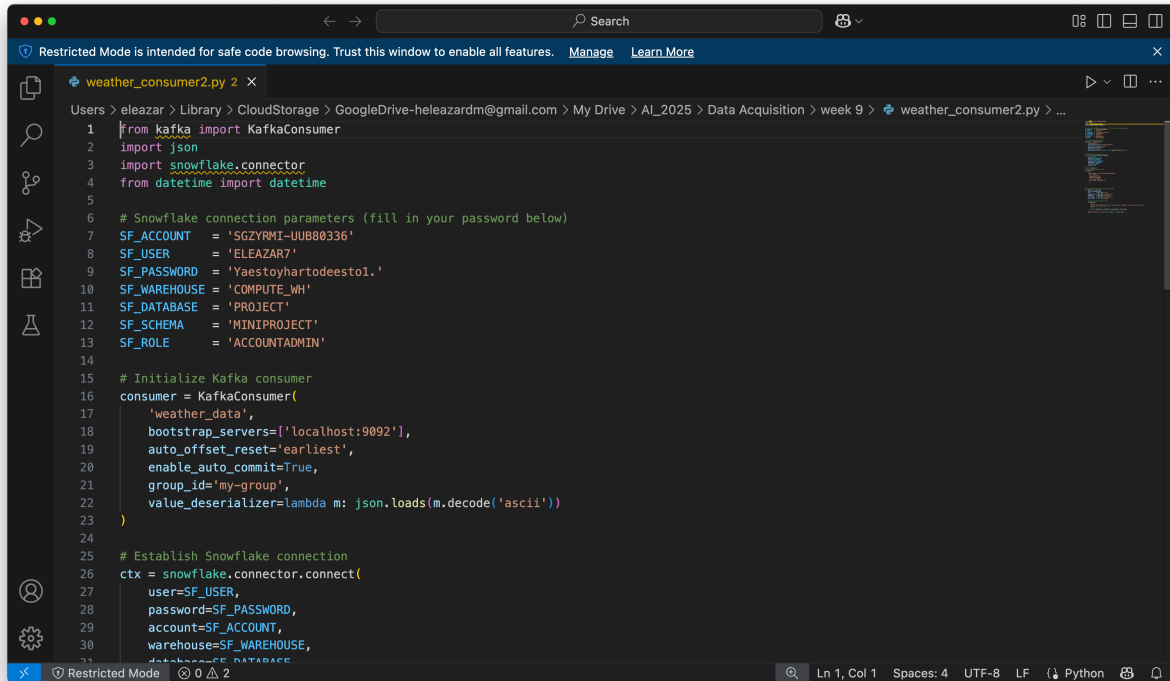
```

Simbolo del sistema - python weather_producer.py
Sending weather data for San Francisco: {'city': 'San Francisco', 'temperature': 6.5, 'humidity': 23, 'wind_speed': 6.5}
Sending weather data for New York: {'city': 'New York', 'temperature': 5.1, 'humidity': 98, 'wind_speed': 3.7}
Sending weather data for San Francisco: {'city': 'San Francisco', 'temperature': 7.7, 'humidity': 58, 'wind_speed': 0.5}
Sending weather data for New York: {'city': 'New York', 'temperature': 2.1, 'humidity': 78, 'wind_speed': 0.7}
Sending weather data for San Francisco: {'city': 'San Francisco', 'temperature': 30.0, 'humidity': 39, 'wind_speed': 6.5}
Sending weather data for New York: {'city': 'New York', 'temperature': -11.0, 'humidity': 25, 'wind_speed': 2.5}
Sending weather data for San Francisco: {'city': 'San Francisco', 'temperature': 15.8, 'humidity': 57, 'wind_speed': 1.5}
Sending weather data for New York: {'city': 'New York', 'temperature': 32.4, 'humidity': 60, 'wind_speed': 13.6}
Sending weather data for San Francisco: {'city': 'San Francisco', 'temperature': 9.8, 'humidity': 19, 'wind_speed': 17.5}
Sending weather data for New York: {'city': 'New York', 'temperature': 11.2, 'humidity': 81, 'wind_speed': 15.5}
Sending weather data for San Francisco: {'city': 'San Francisco', 'temperature': 19.3, 'humidity': 99, 'wind_speed': 8.2}
Sending weather data for New York: {'city': 'New York', 'temperature': 15.5, 'humidity': 48, 'wind_speed': 17.2}
Sending weather data for San Francisco: {'city': 'San Francisco', 'temperature': 9.3, 'humidity': 100, 'wind_speed': 12.4}
Sending weather data for New York: {'city': 'New York', 'temperature': 13.5, 'humidity': 62, 'wind_speed': 13.6}
Sending weather data for San Francisco: {'city': 'San Francisco', 'temperature': 27.1, 'humidity': 51, 'wind_speed': 3.7}
Sending weather data for New York: {'city': 'New York', 'temperature': -2.9, 'humidity': 26, 'wind_speed': 6.5}
Sending weather data for San Francisco: {'city': 'San Francisco', 'temperature': 26.7, 'humidity': 29, 'wind_speed': 8.6}
Sending weather data for New York: {'city': 'New York', 'temperature': 15.9, 'humidity': 17, 'wind_speed': 15.5}
Sending weather data for San Francisco: {'city': 'San Francisco', 'temperature': 26.8, 'humidity': 47, 'wind_speed': 14.6}

Simbolo del sistema - python weather_consumer.py
Received weather data: {'city': 'San Francisco', 'temperature': 3.6, 'humidity': 23, 'wind_speed': 8.9}
Received weather data: {'city': 'New York', 'temperature': 2.8, 'humidity': 23, 'wind_speed': 0.7}
Received weather data: {'city': 'San Francisco', 'temperature': 7.6, 'humidity': 88, 'wind_speed': 18.3}
Received weather data: {'city': 'New York', 'temperature': -19.5, 'humidity': 86, 'wind_speed': 13.5}
Received weather data: {'city': 'San Francisco', 'temperature': 5.5, 'humidity': 58, 'wind_speed': 0.3}
Received weather data: {'city': 'New York', 'temperature': 7.5, 'humidity': 33, 'wind_speed': 1.2}
Received weather data: {'city': 'San Francisco', 'temperature': 7.7, 'humidity': 20, 'wind_speed': 11.2}
Received weather data: {'city': 'New York', 'temperature': 22.3, 'humidity': 5, 'wind_speed': 14.8}
Received weather data: {'city': 'San Francisco', 'temperature': 23.5, 'humidity': 39, 'wind_speed': 9.0}
Received weather data: {'city': 'New York', 'temperature': -12.3, 'humidity': 21, 'wind_speed': 10.1}
Received weather data: {'city': 'San Francisco', 'temperature': 4.2, 'humidity': 25, 'wind_speed': 7.6}
Received weather data: {'city': 'New York', 'temperature': -18.5, 'humidity': 81, 'wind_speed': 9.3}
Received weather data: {'city': 'San Francisco', 'temperature': 21.2, 'humidity': 57, 'wind_speed': 5.7}
Received weather data: {'city': 'New York', 'temperature': 39.2, 'humidity': 3, 'wind_speed': 9.0}
Received weather data: {'city': 'San Francisco', 'temperature': 12.4, 'humidity': 60, 'wind_speed': 8.5}
Received weather data: {'city': 'New York', 'temperature': 20.7, 'humidity': 81, 'wind_speed': 11.5}
Received weather data: {'city': 'San Francisco', 'temperature': 1.6, 'humidity': 19, 'wind_speed': 9.7}
Received weather data: {'city': 'New York', 'temperature': 24.1, 'humidity': 36, 'wind_speed': 11.6}
Received weather data: {'city': 'San Francisco', 'temperature': 15.2, 'humidity': 74, 'wind_speed': 1.4}
Received weather data: {'city': 'New York', 'temperature': 16.0, 'humidity': 99, 'wind_speed': 2.1}
Received weather data: {'city': 'New York', 'temperature': 5.1, 'humidity': 99, 'wind_speed': 3.7}
Received weather data: {'city': 'San Francisco', 'temperature': 7.7, 'humidity': 19, 'wind_speed': 0.5}
Received weather data: {'city': 'New York', 'temperature': 2.1, 'humidity': 78, 'wind_speed': 0.7}
Received weather data: {'city': 'San Francisco', 'temperature': 30.0, 'humidity': 45, 'wind_speed': 6.5}
Received weather data: {'city': 'New York', 'temperature': -11.0, 'humidity': 100, 'wind_speed': 2.5}
Received weather data: {'city': 'San Francisco', 'temperature': 15.8, 'humidity': 43, 'wind_speed': 1.5}
Received weather data: {'city': 'New York', 'temperature': 32.4, 'humidity': 62, 'wind_speed': 13.6}
Received weather data: {'city': 'San Francisco', 'temperature': 9.8, 'humidity': 47, 'wind_speed': 17.5}
Received weather data: {'city': 'New York', 'temperature': 11.2, 'humidity': 82, 'wind_speed': 15.5}
Received weather data: {'city': 'San Francisco', 'temperature': 19.3, 'humidity': 56, 'wind_speed': 8.2}
Received weather data: {'city': 'New York', 'temperature': 15.5, 'humidity': 48, 'wind_speed': 17.2}
Received weather data: {'city': 'San Francisco', 'temperature': 9.3, 'humidity': 73, 'wind_speed': 12.4}
Received weather data: {'city': 'New York', 'temperature': 13.5, 'humidity': 73, 'wind_speed': 8.0}
Received weather data: {'city': 'San Francisco', 'temperature': 27.1, 'humidity': 51, 'wind_speed': 3.7}
Received weather data: {'city': 'New York', 'temperature': -2.9, 'humidity': 26, 'wind_speed': 6.5}
Received weather data: {'city': 'San Francisco', 'temperature': 26.9, 'humidity': 29, 'wind_speed': 8.6}
Received weather data: {'city': 'New York', 'temperature': -9.5, 'humidity': 92, 'wind_speed': 10.1}
Received weather data: {'city': 'San Francisco', 'temperature': 26.7, 'humidity': 41, 'wind_speed': 15.1}
Received weather data: {'city': 'New York', 'temperature': 15.9, 'humidity': 17, 'wind_speed': 15.5}
Received weather data: {'city': 'San Francisco', 'temperature': 26.8, 'humidity': 47, 'wind_speed': 14.6}
  
```

4. Create weather_consumer2.py with Snowflake integration

- Add my Snowflake credentials (user, password, account, etc.).
- I also include a CREATE TABLE IF NOT EXISTS WEATHER_DATA (...) statement for PROJECT.MINIPROJECT.



```

1 from kafka import KafkaConsumer
2 import json
3 import snowflake.connector
4 from datetime import datetime
5
6 # Snowflake connection parameters (fill in your password below)
7 SF_ACCOUNT = 'SGZYRMI-UUB80336'
8 SF_USER = 'ELEAZAR7'
9 SF_PASSWORD = 'Yaestoyhartodeestol.'
10 SF_WAREHOUSE = 'COMPUTE_WH'
11 SF_DATABASE = 'PROJECT'
12 SF_SCHEMA = 'MINIPROJECT'
13 SF_ROLE = 'ACCOUNTADMIN'
14
15 # Initialize Kafka consumer
16 consumer = KafkaConsumer(
17     'weather_data',
18     bootstrap_servers=['localhost:9092'],
19     auto_offset_reset='earliest',
20     enable_auto_commit=True,
21     group_id='my-group',
22     value_deserializer=lambda m: json.loads(m.decode('ascii'))
23 )
24
25 # Establish Snowflake connection
26 ctx = snowflake.connector.connect(
27     user=SF_USER,
28     password=SF_PASSWORD,
29     account=SF_ACCOUNT,
30     warehouse=SF_WAREHOUSE,
31     database=SF_DATABASE,
32     schema=SF_SCHEMA,
33     role=SF_ROLE
34 )
35
36 # Create table if not exists
37 ctx.cursor().execute(
38     f'CREATE TABLE IF NOT EXISTS WEATHER_DATA (
39         city VARCHAR(20),
40         temperature FLOAT,
41         humidity FLOAT,
42         wind_speed FLOAT,
43         timestamp TIMESTAMP
44     )'
45 )
46
47 # Main loop
48 while True:
49     message = consumer.poll(timeout=1.0)
50     if message:
51         for msg in message.items():
52             data = msg.value
53             timestamp = datetime.now()
54             ctx.cursor().execute(
55                 f'INSERT INTO WEATHER_DATA (city, temperature, humidity, wind_speed, timestamp) VALUES ({data[0]}, {data[1]}, {data[2]}, {data[3]}, {timestamp})'
56             )
57             print(f'Loaded into Snowflake: {data} at {timestamp}')
58             time.sleep(1)
59
60 if __name__ == '__main__':
61     main()

```

5. Execute the enhanced consumer

```

cd C:\kafka\scripts
python weather_consumer2.py

```

```

C:\kafka\scripts>python weather_consumer2.py
Traceback (most recent call last):
  File "C:\kafka\scripts\weather_consumer2.py", line 47, in <module>
    cs.execute(
  File "C:\Users\PC\AppData\Local\Programs\Python\Python311\Lib\site-packages\snowflake\connector\cursor.py", line 1134, in execute
    Error.errorhandler_wrapper(self.connection, self, error_class, errvalue)
  File "C:\Users\PC\AppData\Local\Programs\Python\Python311\Lib\site-packages\snowflake\connector\errors.py", line 279, in errorhandler_wrapper
    handed_over = Error.hand_to_other_handler(
  File "C:\Users\PC\AppData\Local\Programs\Python\Python311\Lib\site-packages\snowflake\connector\errors.py", line 334, in hand_to_other_handler
    cursor.errorhandler(connection, cursor, error_class, error_value)
  File "C:\Users\PC\AppData\Local\Programs\Python\Python311\Lib\site-packages\snowflake\connector\errors.py", line 210, in default_errorhandler
    raise error_class(
snowflake.connector.errors.ProgrammingError: 002003 (42S02): SQL compilation error:
Table 'WEATHER_DATA' does not exist or not authorized.

C:\kafka\scripts>python weather_consumer2.py
Loaded into Snowflake: {'city': 'New York', 'temperature': 0.0, 'humidity': 40, 'wind_speed': 16.4} at 2025-07-01 20:52:11.540140
Loaded into Snowflake: {'city': 'San Francisco', 'temperature': 23.0, 'humidity': 0, 'wind_speed': 9.7} at 2025-07-01 20:52:12.589428
Loaded into Snowflake: {'city': 'New York', 'temperature': 3.8, 'humidity': 98, 'wind_speed': 14.7} at 2025-07-01 20:52:16.542287
Loaded into Snowflake: {'city': 'San Francisco', 'temperature': 4.8, 'humidity': 11, 'wind_speed': 11.8} at 2025-07-01 20:52:17.400999
Loaded into Snowflake: {'city': 'New York', 'temperature': -7.5, 'humidity': 4, 'wind_speed': 13.5} at 2025-07-01 20:52:21.559220
Loaded into Snowflake: {'city': 'San Francisco', 'temperature': 1.9, 'humidity': 80, 'wind_speed': 18.2} at 2025-07-01 20:52:22.380634

```

6. Verification of the correct load in Snowflake.

PROJECT / MINIPROJECT / WEATHER_DATA

Table | ACCOUNTADMIN | just now | 14 | 2.5KB

Description

Write a description or [Generate with Cortex](#)

Assigned contacts (PREVIEW)

- Steward
- Support
- Approver

PROJECT / MINIPROJECT / WEATHER_DATA

Table | ACCOUNTADMIN | just now | 14 | 2.5KB

Table Details | Columns | **Data Preview** | Copy History | Lineage

• COMPUTE_WH 25 Rows • Updated just now

	CITY	TEMPERATURE	HUMIDITY	WIND_SPEED	EVENT_TIME
1	New York	0	40	16.4	2025-07-02T03:52:11.540+0000
2	San Francisco	23	0	9.7	2025-07-02T03:52:12.589+0000
3	New York	3.8	98	14.7	2025-07-02T03:52:16.542+0000
4	San Francisco	4.8	11	11.8	2025-07-02T03:52:17.400+0000
5	New York	-7.5	4	13.5	2025-07-02T03:52:21.559+0000
6	San Francisco	1.9	80	18.2	2025-07-02T03:52:22.380+0000
7	New York	21.3	1	16.7	2025-07-02T03:52:26.559+0000
8	San Francisco	6	5	18.1	2025-07-02T03:52:27.490+0000
9	New York	30.9	99	17.2	2025-07-02T03:52:31.563+0000
10	San Francisco	14.5	95	18.7	2025-07-02T03:52:32.432+0000
11	New York	39.8	15	9.6	2025-07-02T03:52:36.565+0000
12	San Francisco	24.7	96	10.4	2025-07-02T03:52:37.367+0000
13	New York	-19.8	13	15.4	2025-07-02T03:52:41.572+0000
14	San Francisco	7.6	41	7	2025-07-02T03:52:41.930+0000
15	New York	1.1	67	4.8	2025-07-02T03:52:46.573+0000

Databases

Worksheets

Search objects

DEMO_DB

GARDEN_PLANTS

LIBRARY_CARD_CATALOG

PROJECT

SNOWFLAKE

SNOWFLAKE_LEARNING_DB

SNOWFLAKE_SAMPLE_DATA

SOCIAL_MEDIA_FLOODGAT...

UTIL_DB

ACCOUNTADMIN

COMPUTE_WH (X-Small)

Share

PROJECT.MINIPROJECT

Settings

Open in Workspaces

1

```
select * from weather_data;
```

Results

Chart

Results

	CITY	TEMPERATURE	HUMIDITY	WIND_SPEED	EVENT_TIME
1	New York	0	40	16.4	2025-07-01 20:52:11.540 -0700
2	San Francisco	23	0	9.7	2025-07-01 20:52:12.589 -0700
3	New York	3.8	98	14.7	2025-07-01 20:52:16.542 -0700
4	San Francisco	4.8	11	11.8	2025-07-01 20:52:17.400 -0700
5	New York	-7.5	4	13.5	2025-07-01 20:52:21.559 -0700
6	San Francisco	1.9	80	18.2	2025-07-01 20:52:22.380 -0700
7	New York	21.3	1	16.7	2025-07-01 20:52:26.559 -0700
8	San Francisco	6	5	18.1	2025-07-01 20:52:27.490 -0700
9	New York	30.9	99	17.2	2025-07-01 20:52:31.563 -0700
10	San Francisco	14.5	95	18.7	2025-07-01 20:52:32.432 -0700

Query Details

Query duration

108ms

Rows

59

Query ID

01bd67a6-0105-d68f-0...

Show more

CITY

New York

30

San Francisco

Ask Copilot